

JIAYI QIAN

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Research Interests:

- (1) Efficient Systems and Architectures for Emerging AI (Agentic, Embodied, and Neuro-Symbolic AI)
- (2) AI-Driven Computer Architecture and Systems Design.

EDUCATION

Georgia Institute of Technology, Atlanta, GA *Aug 2025 - Present*
Ph.D. in Electrical and Computer Engineering, College of Engineering
Advisor: Prof. Tushar Krishna, Prof. Zishen Wan

Georgia Institute of Technology, Atlanta, GA *Aug 2023 - May 2025*
Master of Science in Computer Science, College of Computing
Advisors: Prof. Yingyan (Celine) Lin and Prof. Tushar Krishna

Tsinghua University, Beijing, China *Aug 2019 - Jul 2023*
Bachelor of Science in Electronic Information Science and Technology
Department of Electronic Engineering

PROFESSIONAL EXPERIENCE

Graduate Research Assistant, Georgia Institute of Technology *Aug 2025 - Present*
Sys/Arch Co-Design for AI; AI for Arch/Sys Mentors: Prof. Tushar Krishna, Prof. Zishen Wan

Research Intern, Georgia Institute of Technology *Oct 2023 - May 2024*
Efficient LLM/Diffusion Algorithms Mentor: Prof. Yingyan (Celine) Lin

Teaching Assistant, Georgia Institute of Technology *Jan 2024 - May 2024*
Graduate Algorithms Instructor: Prof. Xiuwei Zhang

PUBLICATIONS (* INDICATES EQUAL CONTRIBUTION)

Systems and Architectures for Emerging AI

ASPLOS '26	Compositional AI Beyond LLMs: System Implications of Neuro-Symbolic-Probabilistic Architectures Zishen Wan, Hanchen Yang, Jiayi Qian , Ritik Raj, Joongun Park, Chenyu Wang, Arijit Raychowdhury, Tushar Krishna
ASPLOS '25	ReCA: Integrated Acceleration for Real-Time and Efficient Cooperative Embodied Autonomous Agents Zishen Wan, Yuhang Du, Mohamed Ibrahim, Jiayi Qian , Jason Jabbour, Yang Zhao, Tushar Krishna, Arijit Raychowdhury, Vijay Janapa Reddi
HPCA '27 Under review	Fast-AIPS: Integrated Acceleration for Efficient Compositional AI Planning Systems Jiayi Qian , Zishen Wan, Zheng Du, Hanchen Yang, Ananda Samajdar, Arijit Raychowdhury, Tushar Krishna
ICML/ISCA '26 Workshops	DyServe: A Workflow-Aware Serving Layer for Agentic Applications Jiayi Qian , Zishen Wan, Hanchen Yang, Chun Tao, Souvik Kundu, Tushar Krishna

HPCA '26	REASON: Accelerating Probabilistic Logical Reasoning for Neuro-Symbolic AI Zishen Wan, Che-Kai Liu, Jiayi Qian , Hanchen Yang, Arijit Raychowdhury, Tushar Krishna
ISPASS '26	Towards System-2 AI: Workloads and Characterizations of Energy-Based Models Hanchen Yang, Jiayi Qian , Zishen Wan, Jingtian Dang, Ziwei Li, Yilun Du, Tushar Krishna
ISPASS '25	Generative AI in Embodied Systems: System-Level Analysis of Performance, Efficiency and Scalability Zishen Wan, Jiayi Qian , Yuhang Du, Jason Jabbour, Yilun Du, Yang Zhao, Arijit Raychowdhury, Tushar Krishna, Vijay Janapa Reddi

AI-Driven Architecture and Systems Design

HPCA '27 Under review	ArchOrchestra: Orchestrating Accelerator Design via Agent-Driven Exploration Jiayi Qian , Hanchen Yang, Souvik Kundu, Vijay Janapa Reddi, Tushar Krishna, Zishen Wan
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Efficient AI Models and Deployment

ICCV '25	Fewer Denoising Steps or Cheaper Per-Step Inference: Towards Compute-Optimal Diffusion Model Deployment Zhenbang Du*, Yonggan Fu*, Lifu Wang*, Jiayi Qian , Xiao Luo, Yingyan (Celine) Lin
NeurIPS '24	AmoebaLLM: Constructing Any-Shape Large Language Models for Efficient and Instant Deployment Yonggan Fu, Zhongzhi Yu*, Junwei Li*, Jiayi Qian* , Yongan Zhang, Dachuan Shi, Xiangchi Yuan, Roman Yakunin, Yingyan (Celine) Lin

TECHNICAL SKILLS

Programming Languages: Python, PyTorch, C/C++, Verilog, CUDA, Shell, MATLAB
Frameworks and Tools: vLLM, MLC-LLM, TensorRT-LLM, Vivado, Docker, Nsight, VTune